

SinhalaJournal-LLM: A Style-Controlled Large Language Model for Diverse Sri Lankan Newspaper Writing

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The rapid expansion of digital journalism in Sri Lanka has increased the demand for grammatically accurate and stylistically consistent Sinhala content. Although recent advancements in Natural Language Processing (NLP) have significantly improved language understanding in high-resource languages [1], low-resource languages such as Sinhala remain underrepresented in large-scale language modeling research. Existing Sinhala NLP studies have primarily focused on language modeling and text classification, including transformer-based approaches such as SinBERT [2] and SinLlama [3]. However, grammar correction for Sinhala remains largely dependent on rule-based systems developed for constrained domains such as OCR post-processing [4]. Moreover, widely used commercial grammar correction platforms do not provide Sinhala support, creating a technological gap.

This research addresses the absence of a context-aware, journalism-oriented Sinhala grammar correction system by proposing a hybrid architecture that integrates rule-based grammatical validation with a fine-tuned Sinhala Large Language Model. Rather than training a foundational model from scratch, the study leverages the pre-trained SinLlama model [3] and adapts it through task-specific fine-tuning for grammatical error detection and correction. The system combines symbolic linguistic constraints with contextual neural reasoning, enabling both structural precision and adaptive correction.

The proposed solution will be evaluated using standard NLP metrics including Precision, Recall, and F1-score, alongside human evaluation to assess fluency and tone preservation. By advancing context-aware grammar correction for a low-resource language, this research contributes to AI-driven language technology development and supports improvements in digital communication and responsible journalism.

Keywords: Sinhala NLP, Grammar Correction, Low-Resource Languages, Large Language Models, Journalism AI

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LIST OF ABBREVIATIONS

API – Application Programming Interface
BERT – Bidirectional Encoder Representations from Transformers
BLEU – Bilingual Evaluation Understudy
DSR – Design Science Research
F1-Score – Harmonic Mean of Precision and Recall
GLEU – Grammar-oriented Evaluation Metric
GPU – Graphics Processing Unit
ITU – International Telecommunication Union
LLM – Large Language Model
LoRA – Low-Rank Adaptation
LREC – Language Resources and Evaluation Conference
NLP – Natural Language Processing
OCR – Optical Character Recognition
PEFT – Parameter-Efficient Fine-Tuning
ROUGE – Recall-Oriented Understudy for Gisting Evaluation
SDG – Sustainable Development Goal
SinBERT – Sinhala Bidirectional Encoder Representation Model
SinLlama – Sinhala-adapted LLaMA Model
TF-IDF – Term Frequency–Inverse Document Frequency

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1 INTRODUCTION

1.1 Background and Context

The rapid growth of digital journalism has significantly transformed news production and dissemination worldwide. According to the International Telecommunication Union (ITU), global internet usage has exceeded 5 billion users, highlighting the dominance of digital platforms in information consumption [5]. In Sri Lanka, increasing internet penetration and smartphone usage have accelerated the shift from print to digital news media. As a result, journalists are required to produce large volumes of content within short publication cycles. Under such constraints, maintaining grammatical accuracy and linguistic consistency in Sinhala digital news content has become increasingly challenging.

Sinhala, spoken by approximately 16 million native speakers, is morphologically rich and syntactically complex, featuring extensive inflection and flexible word order [6]. These linguistic characteristics increase the difficulty of automated grammatical analysis. While transformer-based models have significantly improved contextual language understanding in high-resource languages [1], research shows that low-resource languages remain underrepresented in multilingual training corpora, leading to performance disparities in downstream tasks [7]. This limitation becomes particularly critical in grammar correction tasks that require precise syntactic and contextual modeling.

Existing Sinhala NLP research has made progress in language representation and classification. SinBERT demonstrated improved performance in Sinhala text classification using monolingual pretraining [2], and SinLlama extended LLaMA through vocabulary augmentation and continual pretraining to enhance Sinhala downstream capabilities [3]. However, these models have been evaluated primarily for classification and language understanding tasks rather than grammatical error detection or correction. In parallel, rule-based Sinhala systems developed for OCR and structured document correction provide deterministic rule-based structural validation within template-based domains [4], yet such systems rely heavily on predefined linguistic rules and lack contextual adaptability for free-form journalistic writing.

In contrast, grammar correction systems for high-resource languages increasingly employ transformer-based and sequence-to-sequence architectures capable of context-aware correction generation [1], [8], [9]. Commercial platforms such as Grammarly and Language Tool illustrate the practical deployment of such systems; however, Sinhala is not supported. Consequently, Sinhala users rely mainly on basic spell-checkers that detect surface-level spelling errors without addressing syntactic or contextual grammar issues.

This disparity highlights a clear technological gap. Despite advancements in Sinhala language modeling, there is no dedicated, context-aware grammar correction system optimized for journalism-oriented Sinhala writing. Addressing this gap requires integrating structured grammatical knowledge with contextual neural reasoning.

Accordingly, this research proposes a hybrid Sinhala grammar correction architecture that combines rule-based grammatical validation with a fine-tuned Sinhala Large Language

Model. Rather than training a foundational model from scratch, the study leverages the pre-trained SinLlama model [3] and adapts it through task-specific fine-tuning for grammatical error detection and correction. The integration of symbolic rules and contextual language modeling enables both structural precision and adaptive correction suitable for newsroom environments.

This research aligns with Sustainable Development Goal (SDG) 4 (Quality Education) by enhancing access to Sinhala language support tools, SDG 9 (Industry, Innovation and Infrastructure) by advancing AI-driven language technologies for low-resource languages, and SDG 16 (Peace, Justice and Strong Institutions) by promoting clarity and reliability in public communication. Furthermore, the project aligns with the Artificial Intelligence and Data Science research cluster, specifically within the Natural Language Processing domain.

Therefore, this study focuses on the Sinhala Grammar Checker component within the broader SinhalaJournal-LLM framework.

1.2 Literature Review

The proposed Sinhala Grammar Checker is designed as a hybrid system that integrates rule-based grammatical validation with a fine-tuned Sinhala Large Language Model (SinLlama). The system aims to provide context-aware grammar correction specifically for journalism-oriented Sinhala writing. By combining symbolic grammatical rules with neural contextual modeling, the system seeks to overcome the limitations of purely rule-based or purely neural approaches. The following review examines existing research relevant to transformer architectures, Sinhala language modeling, rule-based validation, and grammar correction systems in both high-resource and low-resource language settings.

Recent advancements in Natural Language Processing (NLP) have been driven primarily by transformer-based architectures. The attention mechanism introduced by Vaswani et al. [1] enabled models to capture long-range contextual dependencies more effectively than traditional sequence models. Building upon this architecture, BERT [8] introduced bidirectional contextual encoding, significantly improving performance in classification and sequence labeling tasks. Large-scale generative models such as GPT further extended these capabilities to text generation and contextual rewriting [10]. These developments established transformer-based models as the dominant paradigm in modern NLP.

To extend these benefits across languages, multilingual models such as XLM-R were developed to learn cross-lingual representations at scale [11]. However, empirical studies show that multilingual training disproportionately favors high-resource languages, resulting in performance disparities for low-resource languages [7]. Tasks requiring fine-grained syntactic understanding, such as grammatical error detection and correction, are particularly sensitive to such representation imbalance.

Within Sinhala NLP research, early work focused on rule-based and statistical methods. De Silva [12] highlighted the scarcity of advanced computational resources and annotated corpora for Sinhala. Subsequent work explored distributional semantic models, including word embedding evaluation for Sinhala [6], improving representation quality but not directly

addressing grammar correction. The introduction of SinBERT [2] demonstrated that monolingual transformer pretraining improves Sinhala text classification performance compared to multilingual baselines [11]. More recently, SinLlama extended the LLaMA architecture through vocabulary augmentation and continual pretraining, further improving Sinhala downstream task performance [3]. These studies confirm the effectiveness of adapting large transformer architectures to Sinhala. However, their evaluations remain focused on classification and general language modeling rather than grammatical error correction.

Parallel research in Sinhala text processing has explored extractive summarization using TF-IDF and TextRank methods [13]. While effective for sentence ranking and content condensation, these approaches do not evaluate grammatical correctness or perform syntactic correction. Similarly, rule-based validation mechanisms have been applied in OCR-based document processing systems to correct recognition errors [4]. Although such systems demonstrate deterministic grammatical validation within constrained domains, they rely heavily on handcrafted linguistic rules and lack contextual adaptability in free-form text.

In high-resource languages, grammar correction research increasingly employs neural sequence-to-sequence architectures and transformer-based models capable of context-aware correction generation [9]. These systems learn correction patterns from annotated corpora and outperform traditional rule-based approaches in handling contextual ambiguities. Commercial platforms such as Grammarly and LanguageTool further demonstrate practical deployment of context-aware grammar correction systems. However, Sinhala remains unsupported in these tools, limiting access to advanced editorial assistance.

The literature therefore reveals three dominant approaches: rule-based grammatical validation, statistical and embedding-based representation learning, and transformer-based contextual modeling. Rule-based systems provide structural precision but lack flexibility. Statistical approaches improve semantic representation but do not address syntactic correction. Transformer-based Sinhala models improve contextual understanding but have not been adapted for grammar correction tasks. Moreover, no prior research integrates Sinhala-specific LLMs with structured grammatical validation mechanisms in a journalism-focused setting.

This review indicates that although Sinhala language modeling has advanced through architectures such as SinBERT and SinLlama, the application of these models to grammar correction remains unexplored. Addressing this limitation requires adapting an existing Sinhala LLM through task-specific fine-tuning and integrating it with symbolic grammatical rules to achieve both contextual reasoning and structural accuracy.

1.3 Research Gap

The proposed research focuses on developing a hybrid Sinhala grammar correction system that fine-tunes the SinLlama model and integrates it with rule-based grammatical validation. While previous sections reviewed advancements in transformer-based NLP and Sinhala language modeling, it is necessary to critically examine whether existing research sufficiently addresses context-aware grammar correction for journalism-oriented Sinhala writing. The following analysis identifies the specific gaps that motivate the proposed system.

Within Sinhala NLP research, transformer-based models such as SinBERT [2] and SinLlama [3] have demonstrated that monolingual pretraining and continual adaptation can significantly improve representation learning and classification performance. However, these models have not been adapted or evaluated for grammatical error detection or correction. Their primary contribution has been in semantic understanding rather than syntactic validation or correction generation. At the same time, rule-based Sinhala systems developed for OCR and document processing [4] demonstrate rule-based structural validation in controlled document domains, but they lack contextual reasoning and scalability for free-form journalistic writing.

This situation reveals a clear gap. High-resource languages benefit from context-aware neural grammar correction systems, while Sinhala systems remain separated between rule-based validation and general language modeling without direct integration for grammar correction. Transformer-based Sinhala models provide contextual understanding but do not enforce structured grammatical rules. Conversely, rule-based systems enforce grammatical constraints but cannot adapt effectively to stylistic variation or contextual ambiguity. No existing solution combines these strengths into a unified grammar correction framework.

In addition, existing Sinhala research has not specifically addressed grammar correction in the context of journalism. News writing requires careful preservation of tone, factual meaning, and stylistic consistency. Purely generative correction may unintentionally alter meaning, while purely rule-based correction may fail to resolve context-dependent errors. To date, no peer-reviewed study has proposed a journalism-oriented Sinhala grammar correction system that integrates a Sinhala-specific Large Language Model with rule-based validation.

Therefore, the main research gap lies in the absence of a hybrid, context-aware Sinhala grammar correction system that leverages existing Sinhala LLM capabilities while incorporating structured grammatical rules. Rather than training a new language model from scratch, there is an opportunity to fine-tune a pre-trained Sinhala LLM such as SinLlama [3] for grammatical error correction and integrate it with rule-based validation mechanisms. Addressing this gap would enable a more practical and reliable grammar correction solution tailored to Sinhala journalistic writing.

This research aims to design and evaluate such a hybrid system, extending Sinhala LLM applications beyond classification toward real-world grammar correction in newsroom environments.

2 OBJECTIVE

2.1 Main Objective

The main objective of this research is to design, implement, and evaluate a hybrid Sinhala grammar correction system for journalistic writing by fine-tuning a pre-trained Sinhala Large Language Model and integrating it with rule-based grammatical validation, achieving a minimum F1-score of 0.80 in grammatical error detection and correction tasks.

2.2 Specific Objectives

To achieve the main objective, the following specific objectives are defined:

1. Development of a Journalism-Oriented Sinhala Grammar Error Dataset

This objective aims to construct a structured dataset consisting of Sinhala news content containing annotated grammatical errors and corresponding corrected forms. Real-world journalistic articles will be collected and manually annotated to identify syntactic, morphological, and agreement-related errors. The dataset will be divided into training, validation, and testing subsets to enable systematic evaluation. The measurable outcome of this objective is the creation of a sufficiently large annotated corpus suitable for fine-tuning and evaluation of the grammar correction model.

2. Design of a Rule-Based Grammatical Validation Module

This objective focuses on developing a rule-based system that encodes core Sinhala grammatical constraints relevant to formal news writing. The module will include validation rules for verb agreement, noun inflection, case markers, and common syntactic inconsistencies observed in journalistic text. The measurable outcome is the implementation of a deterministic validation component capable of detecting predefined grammatical patterns and generating structured error flags. This module will serve as a baseline and as a complementary component in the hybrid architecture.

3. Fine-Tuning of the Pre-Trained SinLlama Model for Grammar Correction

Rather than training a foundational language model from scratch, this objective involves adapting the pre-trained SinLlama model through task-specific fine-tuning using the constructed grammar dataset. The fine-tuning process will focus on grammatical error detection and correction while preserving contextual meaning and journalistic tone. The measurable outcome includes improved performance in grammatical error detection and correction tasks, quantified using evaluation metrics such as Precision, Recall, and F1-score.

4. Integration of Rule-Based Validation with the Fine-Tuned SinLlama Model

This objective aims to design a hybrid architecture that combines the symbolic precision of rule-based validation with the contextual reasoning capabilities of the fine-tuned SinLlama model. The integration strategy will determine how rule-based detection signals inform or constrain neural correction outputs. The measurable outcome is a unified system capable of

producing structurally accurate and contextually appropriate corrections, outperforming standalone rule-based or neural-only baselines.

5. Quantitative Evaluation of System Performance

This objective involves systematically evaluating the proposed grammar correction system using standard Natural Language Processing metrics, including Precision, Recall, and F1-score. The system's performance will be compared against a rule-based-only baseline to demonstrate measurable improvement. A minimum F1-score of 0.80 will be targeted for grammatical error detection and correction tasks. Statistical analysis will be used to validate performance improvements.

6. Qualitative Human Evaluation of Fluency and Tone Preservation

In addition to quantitative evaluation, this objective focuses on assessing correction quality through human evaluation. Sinhala language experts or journalism professionals will review system outputs to evaluate grammatical correctness, fluency, semantic preservation, and journalistic tone consistency. The measurable outcome includes structured evaluation scores and qualitative feedback that validate the practical usability of the system in newsroom contexts.

3 METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology combined with Experimental Evaluation, which is appropriate for building and rigorously validating a novel computational artefact. The core artefact developed in this study is a Sinhala LLM-based Grammar Checker created by fine-tuning the existing SinLlama model rather than training a language model from scratch. The artefact is designed, implemented, and evaluated in a structured cycle consisting of:

1. Problem identification
2. Artefact design and development
3. Controlled experimental validation
4. Performance comparison with baselines

This approach ensures that the contribution is not merely system development, but a validated improvement over existing Sinhala grammar correction approaches.

The experimental component measures linguistic performance improvements using standard NLP evaluation metrics, while the design science component justifies the architectural and algorithmic decisions underlying the system.

3.2 Technical Methods and Algorithms

Base Model Selection and Adaptation

The proposed system builds upon SinLlama, a Sinhala-adapted Large Language Model developed through tokenizer extension and continual pre-training of the LLaMA architecture [3]. Transformer-based architectures have demonstrated superior contextual modelling capabilities compared to earlier sequence models [1], while large-scale pre-trained language models enable efficient transfer learning [10].

Instead of training a foundational language model from scratch, this research applies task-specific fine-tuning to the pre-trained SinLlama model. Fine-tuning leverages transfer learning principles to adapt the model for grammatical error detection and correction, significantly reducing computational requirements while preserving linguistic knowledge [8].

Grammar Correction Strategy

Grammar correction is formulated as a supervised sequence-to-sequence generation task:

Input: Grammatically incorrect Sinhala sentence

Output: Corrected Sinhala sentence

The fine-tuning process minimizes next-token cross-entropy loss while optimizing correction quality. To ensure computational feasibility, Parameter-Efficient Fine-Tuning (PEFT) techniques are applied, specifically Low-Rank Adaptation (LoRA) [14]. LoRA updates only a

subset of attention parameters while freezing the majority of model weights, reducing memory usage and training time.

LoRA is selected because:

- It enables efficient adaptation of large language models [14].
- It reduces risk of catastrophic forgetting.
- It is suitable for low-resource language adaptation scenarios.

Rule-Based Grammatical Validation

To enhance structural reliability, a rule-based grammatical validation module is integrated with the fine-tuned neural model. Rule-based approaches provide deterministic enforcement of grammatical constraints but lack contextual flexibility. Conversely, neural models provide contextual reasoning but may occasionally generate structurally inconsistent outputs.

The hybrid architecture combines both strengths by:

1. Detecting explicit rule violations using predefined grammatical patterns.
2. Refining corrections through contextual neural generation.
3. Validating final outputs against structural rules.

This integration ensures both structural precision and contextual fluency.

3.3 Tools and Platforms with Justification

The system is implemented using the following technologies:

Tool / Platform	Purpose	Justification
SinLlama	Base LLM	Sinhala-adapted decoder LLM
PyTorch	Model training	Industry-standard deep learning framework
HuggingFace Transformers	Fine-tuning & tokenizer handling	Efficient LLM experimentation
PEFT (LoRA)	Parameter-efficient adaptation	Reduces training cost [14]
Google Colab / GPU Server	Training environment	Required for LLM fine-tuning
ROUGE, BLEU	Evaluation metrics	Standard NLP evaluation

Table 1: Tools and Platforms with Justification

These tools are selected based on compatibility with transformer architectures and strong research adoption in NLP studies [1], [8].

3.4 Data Collection and Ethical Considerations

Dataset Preparation

Sinhala grammar correction datasets are constructed using publicly available Sinhala news articles and manually curated error–correction pairs. Due to limited availability of standardized Sinhala grammar datasets [12], the dataset is carefully designed to include morphological errors, subject–verb agreement violations, case marker inconsistencies, and sentence-level syntactic errors.

- Morphological errors
- Subject–verb agreement violations
- Case marker inconsistencies
- Sentence-level syntactic errors

The dataset is divided into training (80%), validation (10%), and testing (10%) sets to ensure unbiased evaluation.

Ethical Considerations

All textual data are collected from publicly available sources. No personal or sensitive information is processed. Human evaluators involved in qualitative assessment provide informed consent. The research adheres to responsible AI principles and ensures that model outputs do not promote harmful or misleading content.

3.5 Validation Strategy and Performance Metrics

Precision, Recall, and F1-score are widely used evaluation metrics in NLP tasks. For text similarity evaluation, ROUGE-based metrics may also be applied where appropriate [15].

A minimum F1-score target of 0.80 is established to demonstrate measurable improvement over baseline approaches.

Baseline Systems

The proposed hybrid model is compared against:

1. Rule-based grammar correction baseline
2. Pre-trained SinLlama without grammar fine-tuning
3. Multilingual transformer model baseline

This comparison ensures that performance improvements are attributable to task-specific fine-tuning and hybrid integration.

Quantitative Metrics

System performance is evaluated using:

- **Precision** – Proportion of correctly identified corrections
- **Recall** – Proportion of grammatical errors detected
- **F1-Score** – Harmonic mean of Precision and Recall
- **GLEU Score** – Evaluation metric for grammatical error correction
- **Perplexity** – Measure of language fluency

These metrics collectively quantify correction accuracy and detection effectiveness.

Qualitative Evaluation

In addition to quantitative evaluation, human reviewers assess:

- Fluency of corrected text
- Preservation of journalistic tone
- Semantic consistency
- Over-correction or meaning distortion

This ensures that improvements are not purely statistical but also practically meaningful.

4 HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed SinhalaJournal-LLM system is designed as a modular, scalable architecture that integrates multiple language processing components within a unified framework. The overall system consists of four primary components:

1. Sinhala Grammar Checker (This component)
2. Style-Adaptive Headline Generator with Image Generation
3. Sinhala Style Rewriter
4. Sinhala News Summarizer

Each component operates independently but shares a common underlying Sinhala Large Language Model infrastructure. While the Sinhala Grammar Checker can function as a standalone service, it also serves as a shared grammatical validation layer for outputs generated by the Headline Generator, Style Rewriter, and News Summarizer.

In this architecture, all generated text responses may optionally pass through the Grammar Checker as a post-processing stage before being returned to the user. This design ensures consistency, structural correctness, and improved linguistic reliability across all system outputs.

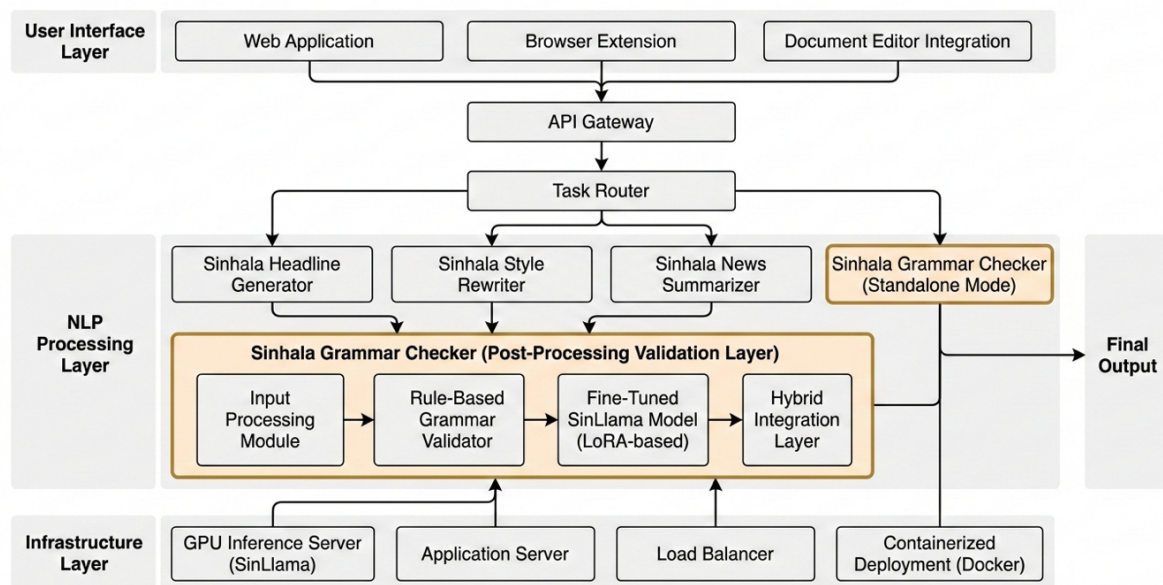


Figure 1: High level system architecture

Figure 1 presents the high-level architecture of the SinhalaJournal-LLM system. The Sinhala Grammar Checker operates both as a standalone service and as a shared post-processing layer

for outputs generated by other NLP components. By combining rule-based validation with a fine-tuned SinLlama model, it ensures consistent grammatical accuracy within a modular, scalable architecture.

4.2 Component Interactions

As shown in Figure 1, the SinhalaJournal-LLM system follows a modular service-oriented architecture where each NLP component operates independently through a centralized API Gateway and Task Router. When a user submits text, the Task Router directs the request to the appropriate component based on the selected task.

For grammar correction requests, input is routed directly to the Sinhala Grammar Checker. For generative tasks such as headline generation, style rewriting, or summarization, the produced output is subsequently routed through the Grammar Checker as a post-processing validation layer before being returned to the user.

Within the Grammar Checker, the Input Processing Module forwards text to both the Rule-Based Grammar Validator and the Fine-Tuned SinLlama model, and their outputs are combined through the Hybrid Integration Layer. This design preserves loose coupling while ensuring consistent grammatical quality across all components.

4.3 Data Flow Explanation

As illustrated in Figure 1, user input first passes through the User Interface Layer and API Gateway. The Task Router determines the requested operation and forwards the text to the selected NLP component.

If the task is grammar checking, the text is processed directly within the Grammar Checker. If the task involves text generation, the generated output is routed to the Grammar Checker as a final validation stage.

Inside the Grammar Checker, the text undergoes normalization, tokenization, and parallel analysis by the Rule-Based Grammar Validator and the Fine-Tuned SinLlama model. The Hybrid Integration Layer merges corrections and produces the final grammatically validated output, which is then returned to the user. This structured pipeline ensures reliability, traceability, and consistent linguistic accuracy.

4.4 Scalability and Security Considerations

The architecture supports scalability through stateless service design, allowing independent horizontal scaling of API services and NLP components. The Fine-Tuned SinLlama model can be deployed on GPU-backed inference servers, while rule-based modules operate on standard application servers. Containerization technologies such as Docker enable flexible and modular deployment.

Security is maintained through HTTPS communication, input validation, authentication controls, and rate limiting. Since textual content is processed without storing personal data, privacy risks are minimized. The centralized Grammar Checker also acts as a quality-control

layer, ensuring that system-generated outputs remain accurate, safe, and linguistically appropriate.

5 USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders of the Sinhala Grammar Checker system are journalists, editors, journalism students, digital content creators, and media organizations that produce Sinhala-language content. These users require accurate and efficient grammar correction tools to improve linguistic quality while maintaining journalistic tone and meaning.

Secondary stakeholders include educational institutions that support Sinhala language learning, software administrators responsible for system deployment and maintenance, and researchers interested in low-resource language technologies. Each stakeholder group interacts with the system at different levels, ranging from end-user text submission to infrastructure management and performance monitoring.

5.2 Functional Requirements

Sinhala Text Input Handling

The system shall accept Sinhala text input through multiple interfaces, including a web application, browser extension, or REST API request. The input module must support Unicode Sinhala characters and allow users to submit paragraphs or full-length news articles. The system shall validate input format before processing to ensure compatibility with downstream modules.

Text Preprocessing

The system shall preprocess the input text prior to grammatical analysis. This includes Unicode normalization, tokenization using the SinLlama tokenizer, sentence segmentation, and removal of extraneous symbols or formatting artifacts. Preprocessing ensures consistent internal representation of text and improves correction accuracy.

Rule-Based Grammar Error Detection

The system shall detect structural grammatical errors using a rule-based validation module. This module shall identify predefined patterns such as subject–verb agreement violations, incorrect case markers, and common newsroom syntactic inconsistencies. Detected rule violations shall be flagged and forwarded to the hybrid integration layer for validation.

Context-Aware Grammar Correction Using Fine-Tuned SinLlama

The system shall generate context-aware grammar corrections using a fine-tuned SinLlama model. The model shall analyze sentence context to detect implicit errors that may not be captured by rule-based validation. Corrections generated by the model shall aim to preserve semantic meaning and journalistic tone while improving grammatical accuracy.

Hybrid Integration of Rule-Based and Neural Outputs

The system shall integrate outputs from the rule-based validator and the fine-tuned neural model using a hybrid decision mechanism. The integration layer shall prioritize structural rule enforcement while allowing context-based refinement from the neural model. Conflicts between rule-based and neural outputs shall be resolved through predefined validation logic.

Generation of Corrected Output

The system shall produce a final grammatically corrected version of the input text. The output shall preserve the original content meaning and formatting where possible. Corrections shall be applied consistently across sentences while maintaining coherence within the full text.

Real-Time Response Capability

The system shall provide grammar correction results within an acceptable response time suitable for newsroom usage. The target response time under normal operating conditions shall be less than two seconds per request for typical news article lengths.

Logging and Monitoring of Requests

The system shall record correction requests, response times, and system performance metrics for monitoring and evaluation purposes. Logs shall support debugging, performance optimization, and research evaluation without storing sensitive user information.

5.3 Non-Functional Requirements

Performance Requirements

- The system shall generate correction results within an average response time of less than 2 seconds under normal operating conditions.
- The system shall achieve a minimum F1-score of 0.80 in grammatical error detection and correction tasks.
- The system shall support concurrent user requests through scalable API deployment.

Security Requirements

- The system shall use secure HTTPS communication for all client-server interactions.
- The system shall implement input validation and sanitization to prevent malicious text injection.
- The system shall restrict API access using authentication mechanisms where required.
- The system shall implement rate limiting to prevent abuse or denial-of-service attempts.

Usability Requirements

- The user interface shall provide clear input and output text areas for correction tasks.
- The system shall display corrected text in a readable and structured format.
- The system shall preserve original formatting wherever possible.
- The interface shall be simple and intuitive for non-technical users such as journalists.

Reliability Requirements

- The system shall maintain stable operation during extended usage periods.
- The system shall handle invalid or malformed inputs without system failure.
- The system shall log system errors for debugging and maintenance.

Scalability Requirements

- The system shall support horizontal scaling through containerized deployment.
- The architecture shall allow independent scaling of the SinLlama inference server and rule-based module.

6 COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The SinhalaJournal-LLM platform targets Sri Lankan digital media organizations, publishing houses, and online news portals that produce high volumes of Sinhala-language content. These organizations require automated support for grammar validation, headline generation, summarization, and style adaptation to improve editorial efficiency and consistency.

Secondary customer segments include:

- Universities and journalism institutes
- Government communication departments
- Digital marketing agencies
- Independent content creators and bloggers

The primary customer persona is a newsroom editor managing multiple daily publications who requires AI-assisted tools to accelerate content production while maintaining linguistic quality.

6.2 Value Proposition

SinhalaJournal-LLM provides an end-to-end AI-assisted newsroom platform that supports:

- Automated grammar correction
- Style-adaptive headline generation

- Controlled tone rewriting
- News summarization

Unlike standalone tools, the integrated architecture ensures that all generated outputs pass through a centralized grammatical validation layer, maintaining linguistic consistency across the platform.

Key measurable benefits include:

- 30–50% reduction in manual proofreading time
- Faster content production cycles
- Improved consistency in journalistic tone
- Reduced editorial workload

The platform addresses a critical technological gap, as no comprehensive AI-powered Sinhala newsroom assistant currently exists.

6.3 Revenue Model

The system follows a Software-as-a-Service (SaaS) model with recurring subscription revenue.

1. Institutional Subscription Plans

Media organizations can subscribe under tiered plans:

- **Basic Plan** – Limited monthly corrections and summarizations
- **Professional Plan** – Higher request limits with priority processing
- **Enterprise Plan** – Unlimited usage, API access, analytics, and dedicated support

Annual subscription packages may be offered at discounted rates.

2. API-Based Pricing

Third-party news platforms, content management systems, and mobile applications can integrate SinhalaJournal-LLM via APIs. Pricing may follow:

- Per-request billing
- Per-token usage pricing
- Custom enterprise API agreements

3. Educational Licensing

Universities and institutes may receive discounted academic licenses to support journalism education and research.

4. Freemium Model

Individual users may access limited monthly usage for free, with optional premium upgrades for advanced features.

6.4 Cost Estimation

The main operational costs include:

- GPU cloud hosting for LLM inference
- API servers and load balancing
- Data storage and monitoring systems
- System maintenance and updates

GPU infrastructure represents the highest recurring cost. However, the use of parameter-efficient fine-tuning techniques (such as LoRA) reduces retraining and computational expenses compared to full model training.

Operational sustainability can be achieved if several mid-scale media organizations subscribe to the Professional Plan. For example, 10 organizations subscribing at mid-tier pricing can cover infrastructure and maintenance costs.

6.5 Pricing Strategy

Pricing must balance affordability within the Sri Lankan market and long-term sustainability.

A possible structure may include:

- **Basic Plan** – LKR 8,000 per month
- **Professional Plan** – LKR 25,000 per month
- **Enterprise Plan** – Custom pricing

Educational institutions may receive discounted pricing. API usage may follow per-request or per-token billing to ensure fairness based on consumption.

The pricing strategy reflects the cost savings compared to hiring additional editorial staff while maintaining competitive affordability.

6.6 Competitive Advantage

SinhalaJournal-LLM offers:

- Dedicated support for Sinhala language
- Journalism-focused optimization
- Hybrid rule-based and LLM-based architecture
- Integration of multiple writing tools in one platform

Most existing solutions either lack Sinhala support or provide limited spell-checking features. This system fills a significant technological gap in the Sri Lankan media industry.

6.7 Intellectual Property Considerations

The intellectual property of the system includes:

- Fine-tuned Sinhala LLM model weights
- Hybrid grammar correction and style adaptation algorithms
- Journalism-specific datasets
- API integration and orchestration framework

The base large language model remains subject to its original licensing terms. Fine-tuned weights, integration logic, and custom-developed modules may be protected under institutional intellectual property policies.

All deployments will comply with open-source licensing requirements and ethical AI usage standards.

7 BUDGET AND JUSTIFICATION

The estimated budget for the SinhalaJournal-LLM project reflects realistic infrastructure requirements for model fine-tuning, experimentation, multi-component deployment, and evaluation.

Since the system integrates multiple NLP components powered by a fine-tuned Sinhala Large Language Model (SinLlama) using Parameter-Efficient Fine-Tuning (LoRA), GPU-based computation is required during experimentation and training phases.

The budget also considers API hosting, system integration, testing, and demonstration deployment to represent the true economic scale of a production-ready AI system.

7.1 Hardware Costs

Item	Estimated Cost (LKR)	Justification
Cloud GPU Rental (NVIDIA T4 / A100 equivalent)	120,000	Approx. 20 USD/hour × 40 hours fine-tuning & experimentation
Development Laptop	0	Existing personal device

Table 2 : Hardware costs

Estimated Hardware Cost: 100,000 – 120,000 LKR

Justification:

Fine-tuning transformer-based models for multiple tasks (grammar correction, headline generation, style rewriting, summarization) requires GPU acceleration. Instead of purchasing a dedicated GPU (which would exceed 300,000 LKR), cloud GPU rental provides a cost-

effective and scalable solution. Using Parameter-Efficient Fine-Tuning (LoRA) significantly reduces memory requirements and training time compared to full model retraining.

7.2 Software Licenses

Software	Cost (LKR)	Justification
Python	0	Open-source
PyTorch	0	Open-source
HuggingFace Transformers	0	Open-source
PEFT (LoRA)	0	Open-source
Docker	0	Open-source
VS Code	0	Open-source

Table 3: Software licenses

Estimated Software Cost: 0 LKR

Justification:

The system relies entirely on open-source technologies, eliminating licensing costs.

7.3 Cloud and Hosting Services

Assumptions:

- 2 vCPU, 4GB RAM instance
- ~10,000 LKR per month
- 6 months usage

Service	Monthly Cost (LKR)	Duration	Total Cost (LKR)
Cloud VM Hosting	10,000	6 months	60,000
Domain Registration	5,000	One-time	5,000

Table 4 : Cloud and hosting cost

Estimated Cloud Cost: 65,000 LKR

Justification:

The cloud infrastructure hosts:

- REST API gateway

- Grammar validation module
- Headline generation module
- Style rewriting module
- Summarization module
- Demo web application
- Monitoring and logging services

Full LLM training does not run continuously. GPU usage is limited to the fine-tuning phase, while inference services operate on optimized deployments.

7.4 Human Effort

Although this is an academic project, the development effort can be estimated to demonstrate the project scale and economic value.

Activity	Estimated Hours
Dataset preparation & annotation	80 hrs
Model fine-tuning & experimentation	120 hrs
System development & API integration	100 hrs
Evaluation & testing	60 hrs
Total Estimated Effort	~360 hours

Table 5 : Human effort

Cost Estimation

Assumptions:

- Intern rate = 1,500 LKR per day
- 1 working day = 8 hours

Total working days:

$$360 \text{ hours} \div 8 \text{ hours/day} = 45 \text{ days}$$

Estimated Human Effort Cost:

$$45 \text{ days} \times 1,500 \text{ LKR/day} = \mathbf{67,500 \text{ LKR}}$$

7.5 Total Estimated Direct Financial Cost

Category	Estimated Cost (LKR)
Hardware (GPU Rental – if required)	0 – 32,000
Software	0
Cloud Hosting & Deployment	38,400 – 43,400
Human Effort (Intern basis)	67,500
Total Estimated Direct Cost	105,900 – 142,900 LKR

Table 6 : Total cost estimation

7.5 Overall Budget Justification

The proposed budget is intentionally minimized through:

- Use of institutional GPU infrastructure
- Adoption of open-source frameworks
- Parameter-Efficient Fine-Tuning (LoRA) to reduce computational overhead
- Limited-duration cloud hosting for deployment and demonstration

The primary financial cost arises from cloud hosting and development effort rather than expensive hardware acquisition.

This approach ensures financial feasibility while maintaining research quality, scalability potential, and technical robustness. The cost structure also demonstrates that SinhalaJournal-LLM is economically viable for both academic research and future commercialization.

8 WORK BREAKDOWN STRUCTURE (WBS)

8.1 Overview of Work Breakdown

The project is divided into the following major phases:

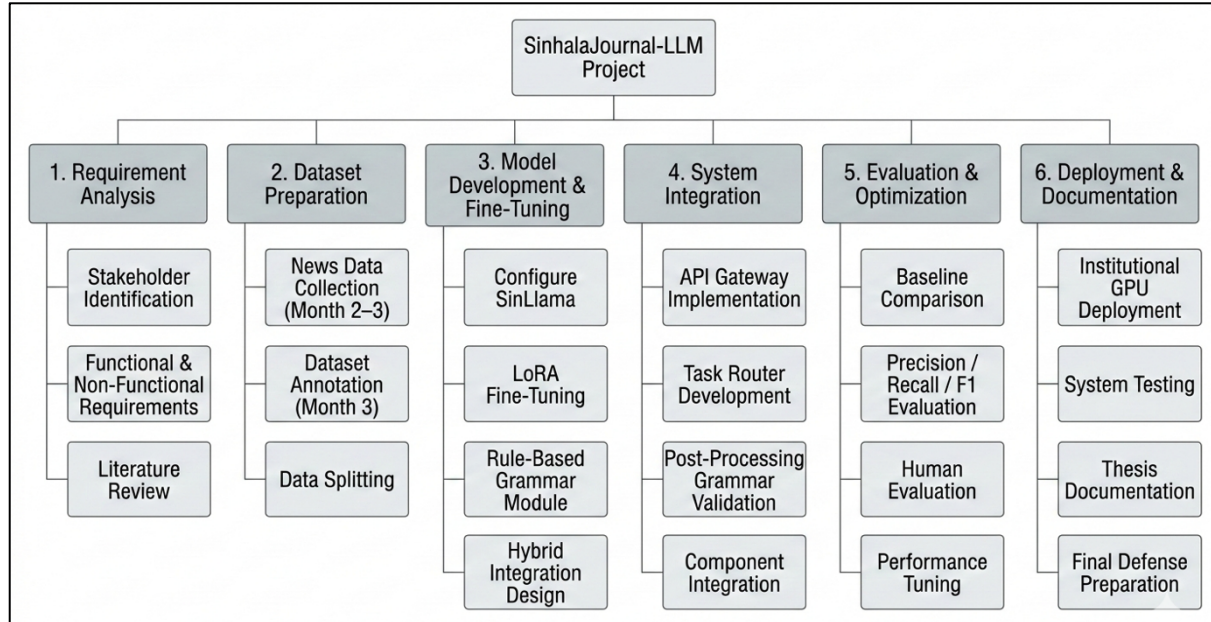


Figure 2 : Work breakdown structure

Figure 2 presents the hierarchical Work Breakdown Structure (WBS) of the SinhalaJournal-LLM project. The project is organized into six major phases with clearly defined tasks to ensure structured development across the one-year timeline. The Sinhala Grammar Checker activities—LoRA fine-tuning, rule-based validation, and hybrid integration—are highlighted to reflect the individual research contribution. The WBS also demonstrates balanced workload distribution and coordinated system-level integration.

8.2 Timeline Alignment

The project spans twelve months, following a phased and progressive development schedule:

- Months 1–2: Requirement analysis and literature review
- Months 2–3: Dataset collection
- Month 3: Dataset annotation
- Months 4–6: SinLlama fine-tuning and rule-based module development
- Months 6–7: Hybrid integration and system integration
- Months 7–9: Experimental evaluation and optimization
- Months 9–10: Institutional deployment and system testing
- Months 7–12: Documentation and thesis writing
- Months 10–12: Final validation and defense preparation

This timeline allows iterative experimentation, refinement, and integration while maintaining feasibility within the academic year.

8.3 Task Breakdown Per Member

Phase	Member 1 – Grammar Checker	Member 2 – Headline Generator	Member 3 – Style Rewriter	Member 4 – News Summarizer
Requirement Analysis	Define objectives and evaluation metrics	Define generation objectives and style criteria	Define transformation objectives and tone categories	Define summarization objectives and evaluation criteria
Dataset Preparation	Prepare annotated error–correction dataset	Prepare labeled headline dataset	Prepare style transformation dataset	Prepare summarization dataset
Model Development	Implement LoRA fine-tuning and rule-based validation	Implement style-conditioned fine-tuning	Implement instruction-based fine-tuning	Implement summarization model
System Integration	Develop API and integrate with router	Develop API and integrate with router	Develop API and integrate with router	Develop API and integrate with router
Evaluation	Evaluate using Precision, Recall, F1, GLEU	Evaluate relevance and style consistency	Evaluate tone preservation and consistency	Evaluate using ROUGE and baseline comparison

Table 7 : Task breakdown per member

8.4 Balanced Workload Distribution

Workload distribution is structured to ensure fairness and technical parity among team members. While the Sinhala Grammar Checker involves both neural fine-tuning and rule-based development, other components require comparable effort in dataset preparation, model adaptation, evaluation, and integration.

Shared tasks such as API implementation, routing logic, and system testing further balance technical contribution. This structured allocation ensures that each member contributes meaningfully to both individual research objectives and overall platform development.

9 GANTT CHART

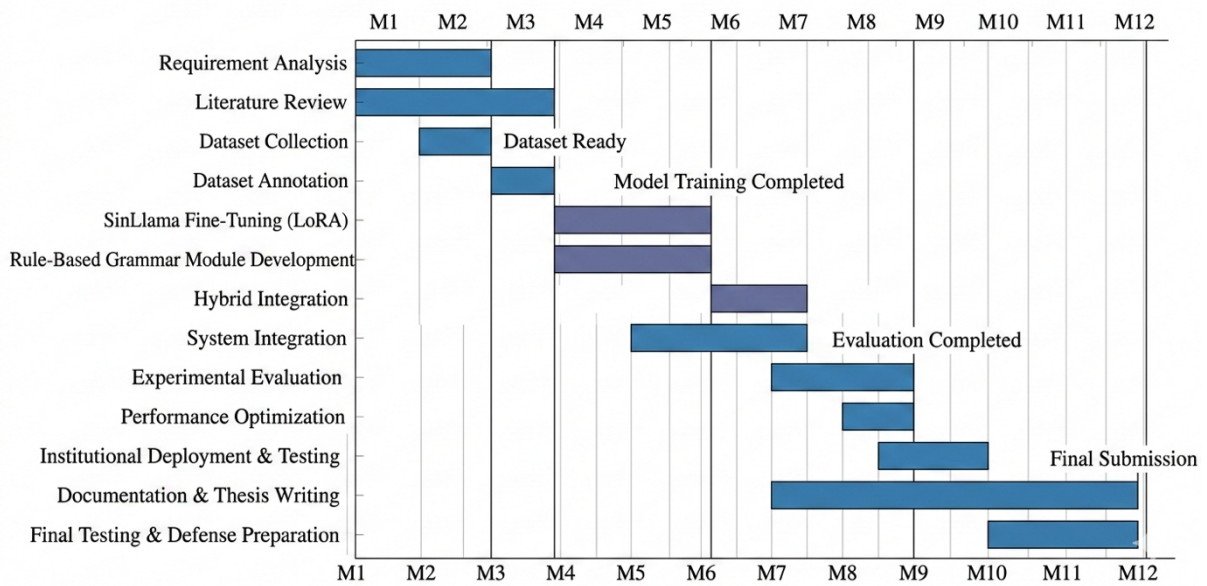


Figure 3 : 12 Month project timeline Gannt chart

Figure 3 presents the 12-month project timeline in the form of a Gantt chart. The schedule aligns research, development, integration, and evaluation phases in a structured and progressive manner. Early months focus on requirement analysis and dataset preparation, followed by model fine-tuning and rule-based module development. System integration and experimental evaluation occur in the middle phase, while deployment, documentation, and final validation are scheduled in the concluding months.

Milestones are clearly defined to monitor progress, including dataset completion, model training completion, evaluation finalization, and final submission. The timeline ensures feasibility, iterative refinement, and sufficient preparation for project defense.

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11 APPENDICES

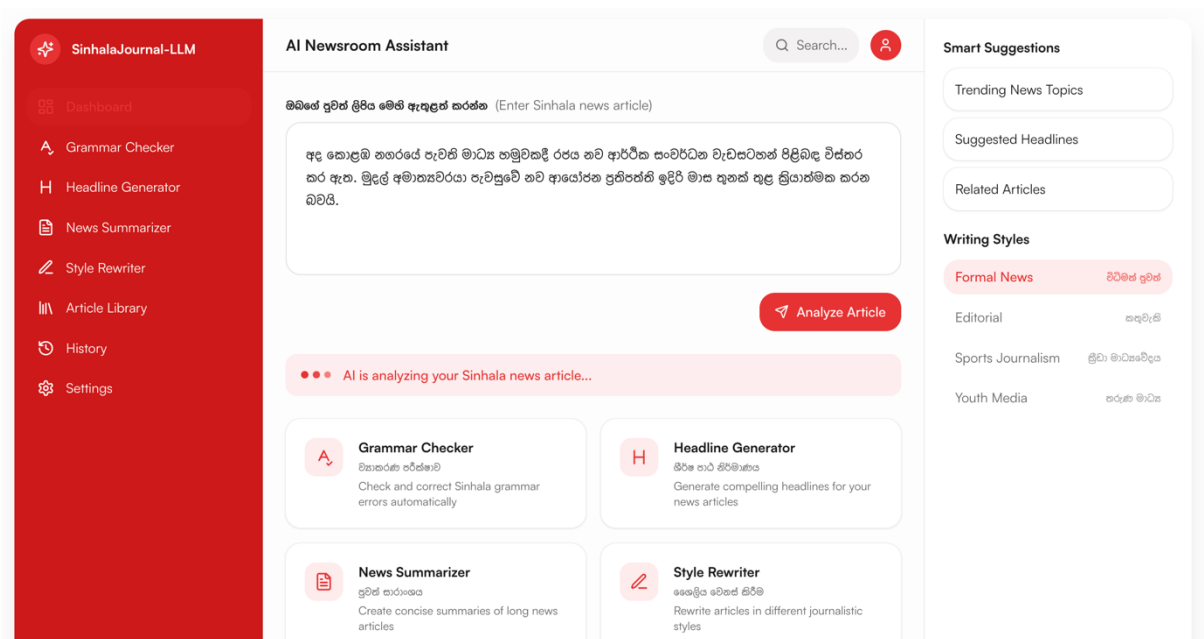
11.1 Risk analysis

The development of the SinhalaJournal-LLM system involves several technical, data, and operational risks. Identifying these risks early helps ensure proper mitigation strategies are implemented during the project lifecycle.

Risk	Description	Mitigation Strategy
Limited Sinhala Data	Sinhala is a low-resource language with limited datasets.	Collect additional news data and use fine-tuning methods like LoRA.
Model Accuracy Issues	The grammar model may produce incorrect corrections.	Use hybrid approach with rule-based validation and human evaluation.
GPU Resource Limitations	Training LLMs requires high computational power.	Use institutional GPU resources and efficient fine-tuning techniques.
Data Quality Problems	Collected news data may contain noise or formatting issues.	Apply preprocessing and manual dataset verification.
System Integration Issues	Integrating multiple NLP components may cause errors.	Use modular architecture with API-based communication.

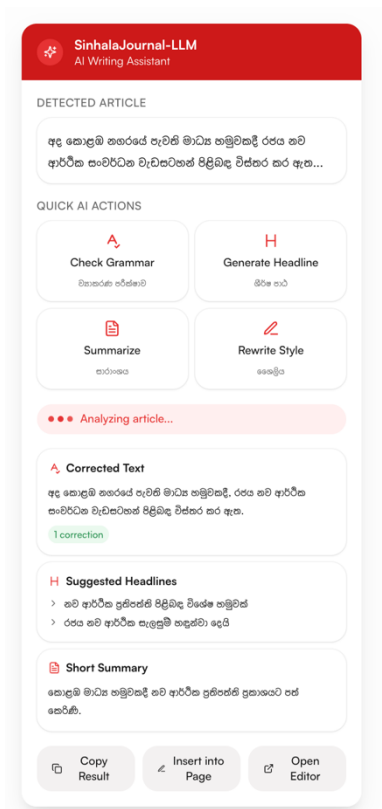
11.2 UI Mockups

Web Application



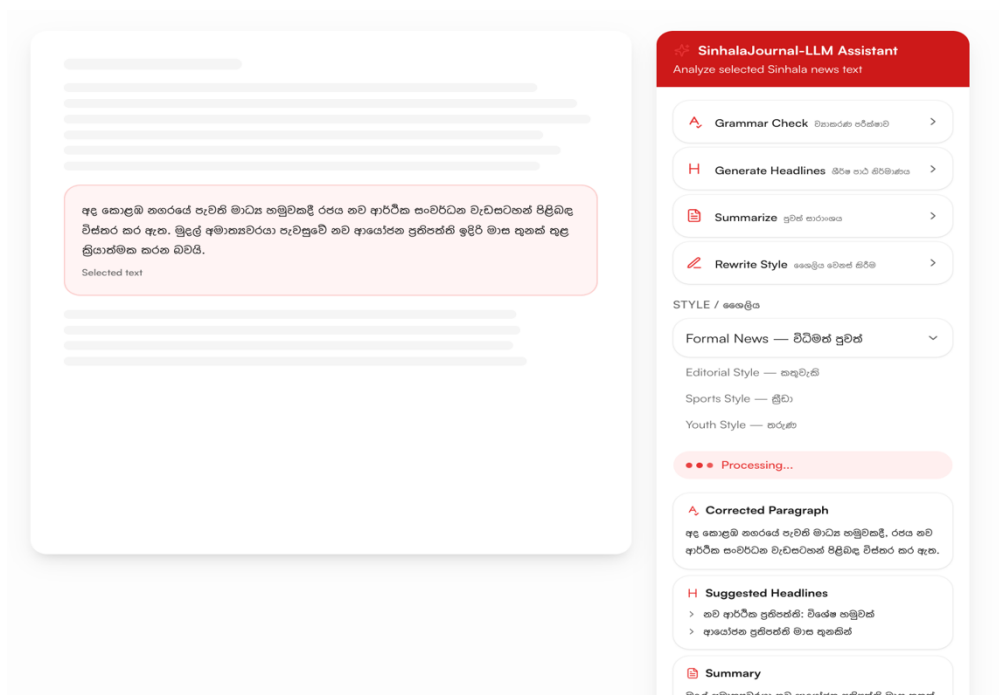
Appendix 1: Ui mockup Web application

Chrome extension



Appendix 2: UI mockup Chrome extension

Google docs integration



Appendix 3: UI mockup Google doc integration

11.3 Ethical Documents



Page 2 of 42 - Integrity Overview

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Appendix 4: Turnitin Report